

$$\langle f, g \rangle = \int_1^5 f(x)g(x) dx$$

a) ON báze  $\text{Span}\{1, x, x^2\}$   
b) DG projekce  $\sin(x)$  na  $\text{Span}\{1, x, x^2\}$  jako  $ax^2+bx+c$   
c)  $\|\sin(x) - (ax^2+bx+c)\|$

a) normalizace  $\vec{v}_1$

$$u_1 = \frac{\vec{v}_1}{\|\vec{v}_1\|} = \frac{1}{\sqrt{\int_1^5 1 \cdot 1 dx}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

\* všechny výpočty jsou zaokrouhlovány na 2 desetinná místa

projekce  $\vec{v}_2$  do  $\text{Span}\{\frac{1}{2}\}$

$$\vec{w}_2 = \left\langle \frac{1}{2}, x \right\rangle \cdot \frac{1}{2} = \int_1^5 \frac{1}{2} \cdot x dx \cdot \frac{1}{2} = 3$$

vypočet  $\vec{u}_2$  + normalizace

$$\vec{u}_2 = \frac{\vec{v}_2 - \vec{w}_2}{\|\vec{v}_2 - \vec{w}_2\|} = \frac{x-3}{\sqrt{\int_1^5 (x-3)(x-3) dx}} = \frac{x-3}{\frac{4\sqrt{5}}{3}} = \frac{3(x-3)}{4\sqrt{5}} = \frac{3\sqrt{5}(x-3)}{12} = \frac{\sqrt{5}(x-3)}{4}$$

projekce  $\vec{v}_3$  do  $\text{Span}\{\frac{1}{2}, \frac{\sqrt{5}(x-3)}{4}\}$

$$\vec{w}_3 = \left\langle \frac{1}{2}, x^2 \right\rangle \cdot \frac{1}{2} + \left\langle \frac{\sqrt{5}(x-3)}{4}, x^2 \right\rangle \cdot \frac{\sqrt{5}(x-3)}{4} = \int_1^5 \frac{1}{2} \cdot x^2 dx \cdot \frac{1}{2} + \int_1^5 \frac{\sqrt{5}(x-3)}{4} \cdot x^2 dx \cdot \frac{\sqrt{5}(x-3)}{4} = \frac{62}{3} \cdot \frac{1}{2} + 8\sqrt{5} \cdot \frac{\sqrt{5}(x-3)}{4} = \frac{21}{3} + 6(x-3) = -\frac{23}{3} + 6x$$

vypočet  $\vec{u}_3$  + normalizace

$$\vec{u}_3 = \frac{\vec{v}_3 - \vec{w}_3}{\|\vec{v}_3 - \vec{w}_3\|} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} = \frac{x^2 - 6x + \frac{23}{3}}{\frac{16\sqrt{5}}{15}} = \frac{15(x^2 - 6x + \frac{23}{3})}{16\sqrt{5}} = \frac{15\sqrt{5}(x^2 - 6x + \frac{23}{3})}{80} = \frac{3\sqrt{5}(x^2 - 6x + \frac{23}{3})}{16}$$

ON báze:  $\text{Span}\{\frac{1}{2}, \frac{\sqrt{5}(x-3)}{4}, \frac{3\sqrt{5}(x^2 - 6x + \frac{23}{3})}{16}\}$

b) projekce  $\sin(x)$  na prostor určený ON bází

$$\begin{aligned} \vec{w} &= \left\langle \frac{1}{2}, \sin(x) \right\rangle \cdot \frac{1}{2} + \left\langle \frac{\sqrt{5}(x-3)}{4}, \sin(x) \right\rangle \cdot \frac{\sqrt{5}(x-3)}{4} + \left\langle \frac{3\sqrt{5}(x^2 - 6x + \frac{23}{3})}{16}, \sin(x) \right\rangle \cdot \frac{3\sqrt{5}(x^2 - 6x + \frac{23}{3})}{16} \\ &= \int_1^5 \frac{1}{2} \cdot \sin(x) dx \cdot \frac{1}{2} + \int_1^5 \frac{\sqrt{5}(x-3)}{4} \cdot \sin(x) dx \cdot \frac{\sqrt{5}(x-3)}{4} + \int_1^5 \frac{3\sqrt{5}(x^2 - 6x + \frac{23}{3})}{16} \cdot \sin(x) dx \cdot \frac{3\sqrt{5}(x^2 - 6x + \frac{23}{3})}{16} \\ &\doteq 0,13 \cdot \frac{1}{2} + (-1,44) \cdot \frac{\sqrt{5}(x-3)}{4} + (-0,13) \cdot \frac{3\sqrt{5}(x^2 - 6x + \frac{23}{3})}{16} \\ &\doteq 0,07 + (-1,44) \cdot \frac{\sqrt{5}x - 3\sqrt{5}}{4} + (-0,13) \cdot \frac{3\sqrt{5}x^2 - 18\sqrt{5}x + 23\sqrt{5}}{16} \doteq 1,59 - 0,32x - 0,05x^2 \end{aligned}$$

c) norma residua

$$\|\sin(x) - (1,59 - 0,32x - 0,05x^2)\| = \sqrt{\int_1^5 (\sin(x) - (1,59 - 0,32x - 0,05x^2))^2 dx} = \sqrt{0,12} \doteq 0,35$$

